

deuteron $\Delta\omega$ mapping via 2-body boundary condition

deuteron (pn) two-body: bound-state scales (frozen)

$B = 2.224566 \pm 0.000001$ MeV
(CODATA mass defect)
 $\Delta\omega = B/\hbar \approx 3.380\text{e}+21$ s⁻¹
 J (2-mode I/F) = $\Delta\omega/2 \approx 1.690\text{e}+21$ s⁻¹
 $1/\kappa$ (tail) ≈ 4.318 fm
 $r_d \approx 2.128$ fm

Square-well example (s-wave)
 $V(r) = -V_0$ for $r < R$, and 0 for $r \geq R$
 $k \cot(kR) = -\kappa$, $\kappa = \sqrt{2\mu B}/\hbar$

Operational I/F for the standing-wave boundary condition.
It does not assert that the nuclear force is literally
a square-well potential.

